



Pitch Control from Broadcast Video: A Computer Vision Pipeline for Set- Piece Analysis

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Executive Summary

This project delivers an open-source tool that turns broadcast football footage into a tactical metric called Pitch Control, giving clubs without expensive tracking systems a way to analyze set pieces using video alone.

The problem. Top clubs use specialist tracking systems to see which team controls each part of the pitch at any moment. Smaller clubs, women's leagues, academies, and scouts cannot afford these systems, so they lose a competitive edge in analyzing corners and free kicks, where spatial control often decides whether a chance is created.

The solution. This project builds a computer vision pipeline that watches broadcast video, identifies each player, tracks them across frames, tells the two teams apart by shirt color, places everyone on a virtual pitch, and calculates how much of that pitch each team controls. Ball position is detected directly from the video. The whole process runs automatically on a normal laptop in about 30 minutes per match clip, with no cloud services or paid software required.

How it was tested. The tool was evaluated on 33 set-piece clips (17 corners, 16 free kicks) from a public research dataset (SoccerNet), and its output was compared frame-by-frame against the dataset's hand-labelled reference data.

What works well.

- The system successfully processed every one of the 33 clips end-to-end.
- For the most operationally relevant question, “who controls the area where the ball is”, the tool's estimate closely matches the reference data (bias = -0.039, histogram overlap = 0.889).
- Global Pitch Control metrics are well-calibrated: `pc_mean bias` = -0.037, `pc_area_gt_0p5 bias` = -0.044.

- The pipeline is fully autonomous: no ground-truth annotations are required at inference time.
- It runs entirely on consumer hardware, so there is no recurring cloud cost.

Where it falls short.

- Inside the crowded penalty box during corners, the system sometimes confuses which team a player belongs to because attackers and defenders stand very close together. This is the main source of error and the clearest target for future improvement.
- Autonomous ball detection produces valid positions for 22 of 33 clips (67% coverage). Clips where the ball is heavily occluded cannot be processed without an alternative position source.

Impact. Clubs and analysts with access to broadcast video and a standard laptop can now generate a credible tactical view of set pieces without buying commercial tracking data. The tool is fully reproducible (locally verified from stored data files), openly documented, and requires no ground-truth annotations at inference time.

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1. Introduction

1.1 Problem Statement

Pitch Control, the probability that a team could reach any point on the pitch first, has emerged as a standard tool for evaluating spatial dominance in elite football, with foundational formulations by Spearman (2018) and Fernández and Bornn (2018). Commercial systems (StatsBomb 360, SkillCorner, Tracab) deliver the underlying tracking data, but their cost restricts access to top-tier competitions.

For the majority of professional clubs, women's leagues, academies, and scouting departments, data-driven set-piece analysis remains out of reach not because of analytical sophistication but because of data access. This project addresses that gap directly, building a pipeline that operates on broadcast footage alone.

1.2 Why Set Pieces

Set pieces (corners and direct free kicks) are tactically high-leverage and analytically tractable. Across 706 set pieces in UEFA Euro 2024, derived from the StatsBomb open-data release (StatsBomb, 2024), 32.4% produced a shot within 10 seconds and 1.8% produced a goal (own analysis, notebook 01). Set pieces are also optimal for a computer vision pipeline: the broadcast camera is near-static, all relevant players are in frame, and ball position can be detected directly from video without requiring an external event feed.

1.3 Stakeholder Impact and Expected Benefits

The primary beneficiaries are organizations that lack access to commercial tracking systems but have broadcast footage available. For these actors, the cost barrier to set-piece analysis is not analytical sophistication but data access. A broadcast-only Pitch Control pipeline removes that barrier directly.

For a **tactical analyst or head coach**, the pipeline provides a spatial view of set-piece control, quantifying which team dominates which zone at the moment of execution, without

requiring a data engineering team or a tracking subscription. The output is a three-panel broadcast overlay that can be interpreted directly on screen without data science expertise.

For a **sporting director or technical lead**, the pipeline enables repeatable, comparable set-piece analysis across opponents and competitions using only publicly available or club-owned broadcast footage. Set-piece patterns can be analyzed for upcoming opponents from broadcast recordings alone.

For a **second-tier club, women's football team, or academy**, the tool removes the hardware and cost barrier entirely. The pipeline runs on a standard laptop in 30 minutes per clip, produces no recurring cost, and requires no proprietary license.

For a **data scientist or analyst building on this work**, the fully documented codebase, stored Parquet outputs, explicit error taxonomy, and two-level reproducibility infrastructure provide a validated foundation for extension rather than a black-box starting point.

1.4 Research Gap

Prior literature covers player detection, tracking, calibration, and Pitch Control modelling individually. Detection has matured through the YOLO family (Redmon & Farhadi, 2018; Jocher et al., 2023); multi-object tracking by association is well-established (Zhang et al., 2022); broadcast camera calibration without ground-truth pitch lines has been addressed by TVCalib (Theiner & Ewerth, 2023); the SoccerNet line of benchmarks (Deliege et al., 2021; Somers et al., 2024) provides annotated broadcast footage; and multi-task learning for joint re-identification, team affiliation, and role classification has been proposed for sports tracking (Mansourian et al., 2023). The end-to-end chain, broadcast pixels through to a distributionally validated tactical metric, without proprietary tracking and without ground-truth annotations consumed at inference time, remains underdeveloped.

1.5 Contribution

This project delivers:

- A reproducible, validated pipeline from broadcast video to Pitch Control, operating without GT annotations at any inference step
- Fully autonomous operation: player detection, ball detection, camera calibration, and team assignment all run from broadcast video alone
- Distributional validation against open SoccerNet GSR annotations (Somers et al., 2024) with explicit bias attribution per metric
- ICC-based effective sample size analysis that quantifies the binding statistical constraint and informs future cohort design
- Consumer-hardware execution (Apple Silicon MPS or equivalent) with no cloud dependency

1.6 Research Scope and Boundaries

Data scope: The primary validation dataset is SoccerNet GSR 2024 (Somers et al., 2024): 33 clips covering two set-piece classes (17 corners, 16 direct free kicks). StatsBomb Euro 2024 open data (StatsBomb, 2024) is used only for distributional context on player counts and set-piece outcomes; it is not part of the primary validation.

Set-piece scope: Only corners and direct free kicks are analyzed. These were selected because the broadcast camera is near-static for both, all relevant players are in frame at the moment of execution, and ball position is reliably detectable from video. Throw-ins, goal kicks, indirect free kicks, and open-play sequences are out of scope.

Out of scope: Player re-identification across clips, multi-camera setups, non-broadcast footage, and tracking over full match sequences are excluded. The pipeline is validated as a static-frame set-piece tool, not a real-time tracking system.

1.7 Research Structure and Preview

Section 2 defines the research objectives and success criteria. Section 3 lays out the project timeline, milestones, and constraints. Section 4 describes the full pipeline architecture and technology stack. Section 5 explains the CRISP-DM methodology and its adaptations. Section 6 documents each phase of development in detail. Section 7 discusses the findings, identifies cross-metric patterns, and articulates methodological limits. Section 8 draws conclusions, proposes a development roadmap, and outlines future work. Appendices provide the project folder structure, model parameters, data sources, and full reproducibility instructions.

2. Objectives

2.1 Primary Objective

Develop a reproducible, fully autonomous computer vision pipeline that extracts Pitch Control from broadcast set-piece frames and produces distributions comparable to ground-truth annotation-derived distributions, using only open-source tools and consumer hardware.

2.2 Research Questions

- **RQ1:** Can a broadcast-video-only pipeline produce Pitch Control distributions comparable to GT-annotation distributions for set-piece frames?
- **RQ2:** What is the dominant source of systematic bias in pipeline-derived Pitch Control, and is it attributable to specific pipeline components?
- **RQ3:** Which Pitch Control summary metrics are most robust to broadcast-pipeline noise?

2.3 Academic Objectives

- Demonstrate end-to-end distributional validation methodology for a broadcast computer vision pipeline, moving beyond aggregate accuracy to per-metric bias attribution.
- Establish which Pitch Control summary metrics are robust to broadcast-pipeline noise and which are structurally contaminated by specific component failures.
- Contribute empirical evidence on the limits of color-based team assignment in crowded set-piece environments, informing future work on supervised team-classification approaches.
- Quantify within-clip temporal correlation via ICC analysis to characterise the effective statistical power of the validation cohort.
- Produce a reproducible codebase and validated outputs that enable independent replication and extension.

2.4 Practical Objectives

- Deliver a working, end-to-end pipeline executable on consumer hardware in under 45 minutes on Apple Silicon MPS (longer on CPU-only hardware) for 33 clips, without cloud infrastructure or proprietary data.
- Identify which PC metrics are operationally deployable by clubs without access to GT tracking, and which require further development before use.
- Produce broadcast-overlay visualizations (three-panel stills and animated GIFs/MP4s) that allow a tactical analyst to interpret pipeline outputs directly on the broadcast frame.
- Document data-quality issues in SoccerNet GSR (notably the `action_position` global-frame-number misinterpretation) to inform future dataset users.

2.5 Expected Outcomes and Deliverables

Technical deliverables:

- Detection parquets (detections_soccana_tvcalib.parquet, detections_gt_full.parquet) for all 33 clips
- Ball position cache (ball_positions.parquet) and pitch control surfaces (pitch_control_soccana_tvcalib.parquet, pitch_control_gt_full.parquet)
- Validation outputs (validation_summary_tvcalib.parquet, validation_paired.parquet, KS table figure)
- ICC analysis outputs (icc_per_metric.parquet)
- Broadcast-overlay visualizations: static three-panel stills, animated GIFs, and MP4 clips for representative corner and free-kick sequences

Academic deliverables:

- Validated distributional comparison of pipeline-derived vs GT-derived Pitch Control on 22 clips with autonomous ball detection
- Mechanistic bias attribution identifying three distinct error regimes: global underestimation, sign inversion, and calibrated range
- ICC (2,1) analysis quantifying effective sample size and within-clip correlation structure
- Documented methodology for adapting a dynamic Pitch Control model to a zero-velocity set-piece context

2.6 Success Criteria

Table 1. Project success criteria and targets.

Criterion	Target
End-to-end execution	All 33 clips, consumer hardware, <45 min (MPS) / <90 min (CPU)
Bias	$ \text{bias} < 0.10$ on <code>pc_at_ball</code>
Histogram overlap	>0.80 on at least one metric
Bias explanation	Mechanistic, attributable to specific component
Reproducibility	Full pipeline re-runnable from raw SSD data

2.7 Ethical Considerations

Data access: SoccerNet GSR requires a credentialed download via the SoccerNet API; credentials were obtained through the standard academic access request process. StatsBomb open data is freely available under CC BY-SA 4.0. Soccana model weights are publicly hosted on HuggingFace under an open license.

No personal data: No individual player identities, biometric data, or personally identifiable information are stored, processed, or published. Player positions are treated as anonymous spatial coordinates.

Model weights: The Soccana YOLOv11n weights are used as-is from HuggingFace; their original license terms apply and no weights are redistributed.

Reproducibility and transparency: All pipeline code, parameter choices, and known data-quality issues (including the `action_position` misinterpretation) are documented and stored in the project folder. No results are selectively reported; the full validation table including unfavorable metrics is published.

3. Project Timeline

3.1 Planning Approach

The project follows CRISP-DM's phased structure, with explicit feedback loops between Data Understanding and Data Preparation to accommodate mid-project corrections. The plan below is presented in illustrative weeks (W1 through W11) rather than calendar dates, reflecting the iterative nature of the work and the fact that several phases overlap.

3.2 Phase Plan

Table 2. CRISP-DM phase plan with milestones and deliverables.

Week	CRISP-DM phase	Activity	Deliverable
W1-W2	Business Understanding	Problem framing, stakeholder identification, set-piece literature review	Define Research questions
W2-W4	Data Understanding	StatsBomb Euro 2024 EDA, SoccerNet GSR scan, action_position audit	setpieces.parquet, gt_spatial_benchmarks.parquet
W3-W5	Data Preparation (pipeline track)	TVCalib integration, Soccana detection + ByteTrack, team assignment design	homographies_tvcalib.parquet, detections_soccana_tvcalib.parquet
W4-W5	Data Preparation (GT track)	SoccerNet bbox_pitch parsing, autonomous ball detection integration	detections_gt_full.parquet, ball_positions.parquet
W5-W7	Modeling	Laurie Shaw TTI zero-velocity adaptation, PC surface computation	pitch_control_.parquet
W7-W9	Evaluation	Distributional KS, paired Pearson/MAE/bias, ICC analysis	validation_summary_tvcalib.parquet, validation_paired.parquet, icc_per_metric.parquet
W9-W10	Deployment	Broadcast-overlay visualizations, three-panel stills, animated GIF/MP4	still_*.png, anim_*.gif, video_*.mp4
W10-W11	Reporting	Notebook narrative, thesis write-up, final review	report.md, executable notebooks

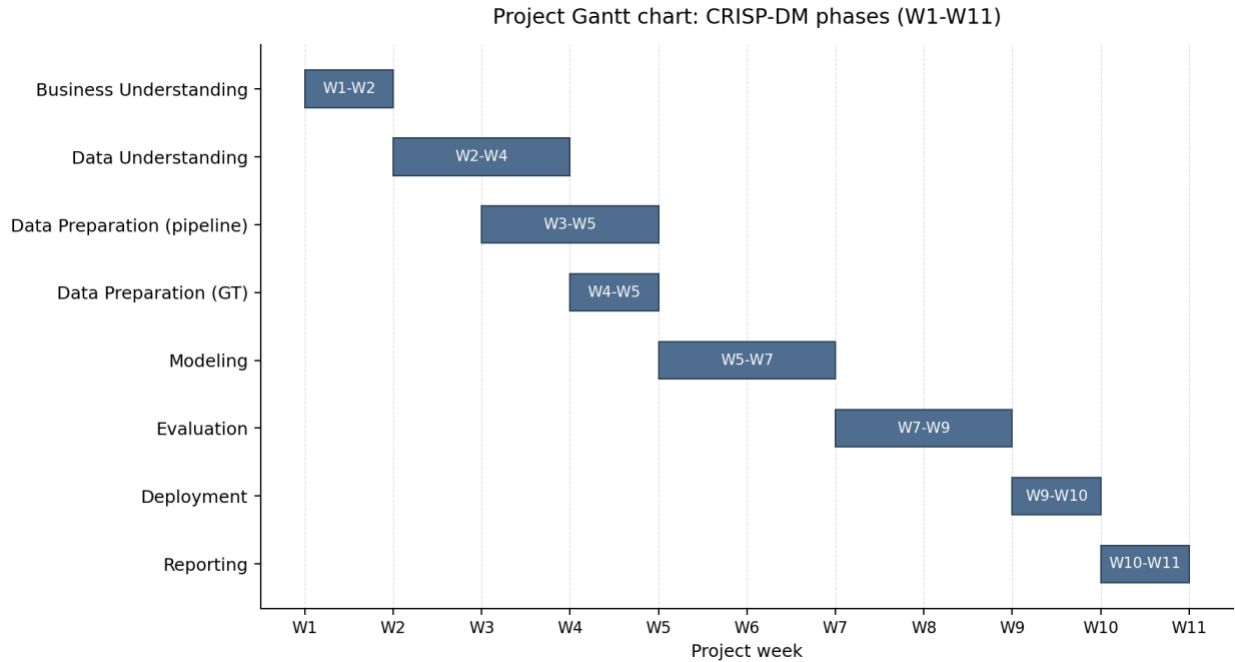


Figure 1. Project Gantt chart showing the eight CRISP-DM phases across illustrative weeks W1-W11. Overlapping bars highlight that the pipeline and GT data-preparation tracks run partly in parallel.

3.3 Key Milestones

1. **Research scope locked** (end W2): primary validation dataset, set-piece classes, success criteria all defined.
2. **TVCalib batch complete** (W4): 1,023 homographies computed for all 33 clips, removing GT pitch-line dependency at inference.
3. **action_position data-quality discovery** (W4-W5): mid-project finding that action_position is a global broadcast frame number, not a clip-local index. Required rewriting the frame-window logic before modelling could produce correct outputs.
4. **First end-to-end pipeline run** (W6): pipeline runs all 33 clips end-to-end with zero homography failures.
5. **Validation cohort finalized at 22 clips** (W7): the effective PC validation set is 22 clips, reflecting autonomous ball detection coverage (67%). SNGS-125 and SNGS-145 lack GT annotations in frames 1-31; a further 9 clips lack sufficient autonomous ball detections in the critical early frames.

6. **KS validation table complete** (W8): per-metric distributional comparison published with no selective reporting.
7. **Error taxonomy and ICC analysis complete** (W8–W9): three-regime error partition (global underestimation, sign inversion, calibrated) and within-clip ICC values (0.83–0.92) with effective sample sizes of 24–26 observations.
8. **Final deliverables completed** (W11): Parquet outputs, executable notebooks, thesis report, and overlay visualizations.

3.4 Constraints and Dependencies

Constraints:

- *Data access*: SoccerNet GSR requires a credentialed download; access was obtained through the academic process. StatsBomb open data is freely available.
- *Time limitations*: The Sports Data Campus submission deadline of 30 June 2026 fixes the upper bound of the planning window and is the binding driver of the eleven-week schedule shown in Table 2 and Figure 1.
- *Hardware*: Apple Silicon laptop with 16 GB unified memory; SoccerNet video (~35 GB) hosted on an external USB-C SSD.
- *External component*. TVCalib (Theiner & Ewerth, 2023) runs in a sibling conda environment with PyTorch 2.x patches; required configuring and validating a research codebase before pipeline integration could proceed.

Dependencies:

- Detection requires TVCalib homographies (or at minimum a valid camera model) before pixel-to-pitch projection is meaningful.
- Pitch Control computation requires both detections and ball positions per frame.
- Validation requires aligned pipeline and GT Pitch Control surfaces on a shared clip set.
- Notebooks 02 through 04 are executable from committed Parquet outputs, allowing reproduction without the SSD or TVCalib environment after the initial pipeline run.

3.5 Business Rules

A small number of explicit project rules shaped execution from start to finish. They are documented here so the planning context is auditable:

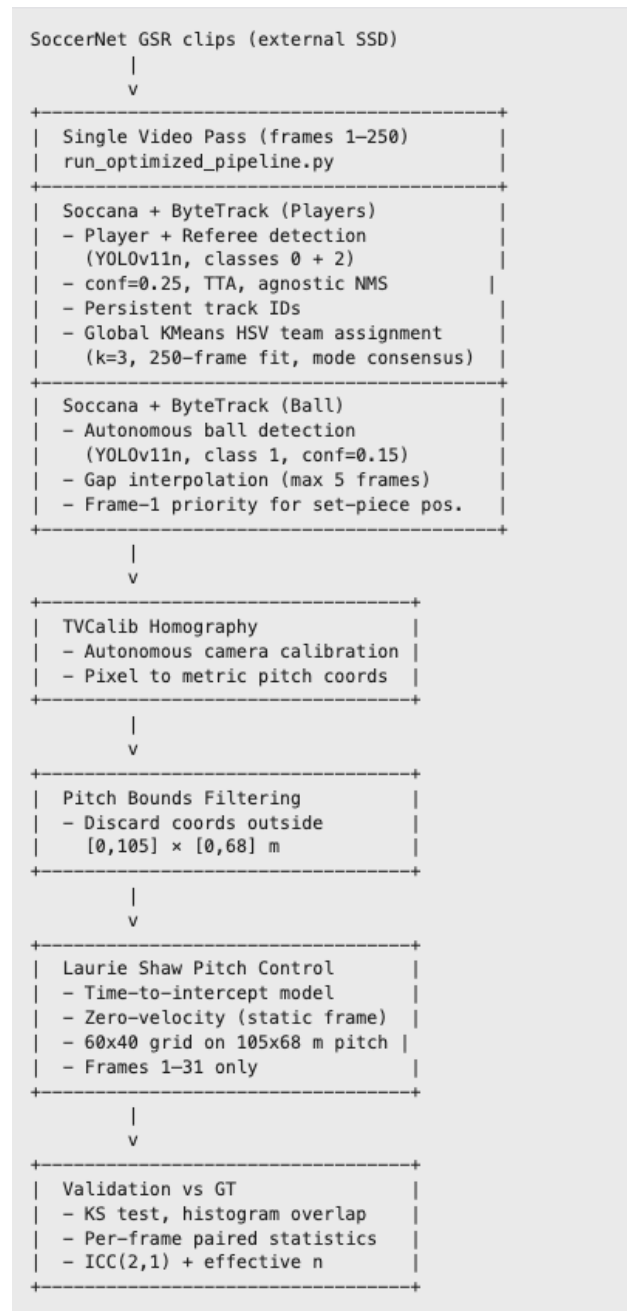
- *Inference-time data-leak rule*: SoccerNet GSR ground-truth annotations may be used only for validation. They are never consumed at detection, tracking, calibration, or ball detection inference time. This rule is what makes the pipeline a defensible broadcast-only system and is the central methodological constraint of the project.
- *Data licensing*: StatsBomb Euro 2024 open data is used under CC BY-SA 4.0. SoccerNet GSR is accessed through the credentialed academic process (Somers et al., 2024). Soccana weights are used under their published HuggingFace open licence; no weights are redistributed.
- *Reproducibility rule*: The pipeline must execute end-to-end on a consumer laptop with no cloud dependency. All intermediate outputs are stored as Parquet so that notebooks 02 through 04 reproduce on a fresh copy of the project folder without the SSD or TVCalib environment.
- *Ethics rule*: No personal data, biometric data, or player identities are stored, processed, or published. Player positions are treated as anonymous spatial coordinates throughout (full statement in Section 2.7).

3.6 Risk and Mitigation

The most material risk that materialized was the `action_position` misinterpretation: the pipeline was initially processing end-of-clip open-play frames instead of set-piece formations. This was caught during qualitative inspection of intermediate outputs in W4–W5, fixed by computing the center frame as `FRAME_WINDOW + 1 = 16` rather than treating `action_position` as a clip-local index, and documented as a methodological contribution. The mitigation lesson is that intermediate visual inspection is essential when the pipeline produces metrics that look plausible in aggregate but are semantically wrong.

4. Conceptual and Technological Architecture

4.1 Pipeline Overview



4.2 Technologies

Table 3. Pipeline components and underlying technologies.

Component	Technology
Detection	Soccana YOLOv11n, football-finetuned, 2.6M params (Jocher et al., 2023)
Tracking	ByteTrack (Zhang et al., 2022) via Ultralytics
Team assignment	Global KMeans (k=3) on track-mean HSV + cross-frame mode consensus
Calibration	TVCalib (Theiner & Ewerth, 2023)
Ball detection	Autonomous: Soccana class=1, conf=0.15, ByteTrack + gap interpolation
Pitch Control	Time-to-intercept model (Shaw, 2020)
Data	SoccerNet GSR 2024 (Somers et al., 2024); StatsBomb open data (StatsBomb, 2024)
Language	Python 3.11
Package management	uv + pyproject.toml + uv.lock
Hardware	Apple Silicon (MPS), or any CUDA/CPU-capable host

4.3 Coordinate Systems

Table 4. Coordinate-system conventions used across data sources and the pipeline.

System	Convention
StatsBomb	120 yd x 80 yd, origin top-left
Pipeline / mplsoccer	105 m x 68 m, origin top-left
SoccerNet GSR bbox_pitch	centered origin (+-52.5 m, +-34 m)

Conversions: GSR to pipeline: $x = x_{\text{gsr}} + 52.5$, $y = y_{\text{gsr}} + 34$

StatsBomb to pipeline: $x = x_{\text{sb}} \times (105/120)$, $y = y_{\text{sb}} \times (68/80)$

4.4 Core Libraries and Tools

Table 5. Core Python libraries and their roles in the pipeline.

Library	Version	Role
ultralytics	8.3.107	Soccana detection + ByteTrack tracking
torch	>=2.1.0	Inference backend (Apple MPS)
opencv-python	>=4.8.0	Frame I/O, HSV colour extraction, video rendering
pandas	>=2.1.0	Tabular data pipeline
numpy	>=1.26.0	Array operations, homography math
scipy	>=1.11.0	KS test, spatial distance computations
scikit-learn	>=1.3.0	KMeans team assignment
mplsoccer	>=1.4.0	Pitch visualisation
pyarrow	>=14.0.0	Parquet I/O
statsbombpy	>=1.14.0	StatsBomb open data access
huggingface-hub	>=0.20.0	Soccana weight download
pingouin	>=0.5.0	ICC (2,1) computation and effective sample size
python-dotenv	>=1.0.0	SSD path and credential configuration

TVCalib (Theiner & Ewerth, 2023) runs in a separate sibling conda environment with its own dependencies (PyTorch 2.1, kornia 0.8.2, pytorch-lightning 2.6.1) and is invoked via subprocess from `run_tvcalib_batch.py`.

4.5 Development and Hardware Environment

- **Hardware:** Apple Silicon laptop (M-series, 16 GB unified memory), MPS backend for PyTorch inference. The pipeline also runs on CUDA or CPU; detection will be slower on CPU-only hardware.
- **Storage:** SoccerNet GSR video data (approx. 35 GB) on an external USB-C SSD. All intermediate outputs (Parquet files, figures) live in the repository `outputs/` directory and are committed; video frames are not committed.
- **Software:** macOS 15, Python 3.11 (uv + pyproject.toml). Detection runs on MPS; Pitch Control computation and validation run on CPU.

- **Execution profile:** approx. 30 min total for 33 clips (TVCalib batch around 14 min, Soccana detection around 15 min, all downstream scripts <1 min combined).

4.6 Scalability and Constraints

Horizontal scaling: The pipeline processes clips sequentially; parallelization across clips is straightforward but was not implemented, as the 30-minute runtime is acceptable for the 33-clip cohort.

TVCalib coupling: Homography computation requires TVCalib in a sibling directory with its own environment. Pre-computed homographies are committed to the repository (`homographies_tvcalib.parquet`), decoupling all downstream scripts from the TVCalib dependency for normal operation.

Ball detection: Autonomous ball detection produces valid positions for 22 of 33 clips. The 11 remaining clips lack sufficient ball detections in the critical early frames due to occlusion or ball out of frame. The effective PC validation set is therefore 22 clips.

Memory: Peak memory during detection inference is approximately 4 GB (MPS), dominated by the YOLO model. Pitch Control computation is CPU-bound and uses <1 GB.

5. Methodology: CRISP-DM

CRISP-DM (Cross-Industry Standard Process for Data Mining; Chapman et al., 2000) is a six-phase framework that organizes a data project into Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. Each phase has explicit inputs, outputs, and a feedback path to earlier phases, which makes the framework well-suited to projects where validation findings can require revisiting data assumptions. This section explains why CRISP-DM was selected for this project, summarizes how each phase maps to the repository's notebooks and scripts, and notes two methodology adaptations needed for the broadcast computer-vision context.

5.1 Why CRISP-DM

CRISP-DM was selected as the organizing framework for three reasons specific to this project:

Iterative fit: Pipeline development rarely proceeds linearly. The discovery that `action_position` in SoccerNet GSR is a global broadcast frame number (not a clip-local index), uncovered during data understanding, required revising the frame-window logic before preparation and modelling could proceed correctly. CRISP-DM's explicit feedback loop between phases accommodates this kind of mid-project revision without treating it as a failure.

Validation emphasis: The evaluation phase in CRISP-DM is a first-class phase, not an afterthought. For this project, where the central research question is whether a broadcast pipeline produces distributions comparable to ground truth, the evaluation phase is where the research question is answered.

Reproducibility: CRISP-DM's phase separation maps directly onto the script/notebook structure of the project folder: each phase has identifiable inputs, processing steps, and stored Parquet outputs. A future user can enter the pipeline at any phase using stored intermediate outputs, verifying reproducibility independently.

5.2 Phase Summary

This subsection maps the CRISP-DM phases to the concrete artefacts of the project: every phase has identifiable inputs, a script or notebook that performs the work, and stored output. The table is intended to function as both a methodology summary and an entry point for reproducing any individual phase.

Table 6. CRISP-DM phase to notebook / script and committed output mapping.

Phase	Notebook / Script	Key Output
Business Understanding	nb01	Problem framing, stakeholder identification, set-piece EDA
Data Understanding	nb01	StatsBomb EDA, SoccerNet GSR scan, action_position audit
Data Preparation	run_tvcalib_batch, run_optimized_pipeline, dump_gt_setpieces	Detection, ball-position, and team-assignment Parquets
Modeling	nb02, run_pc_*	Pitch control surfaces, summary metrics
Evaluation	nb03, ks_table_tvcalib.py, compute_icc.py	Validation tables, distributional + paired statistics, ICC
Deployment	nb04, render_*	Broadcast-overlay visualizations, GIFs, MP4s

5.3 Methodology Adaptations

Two adaptations were required to apply CRISP-DM to this computer vision pipeline context.

Static-frame modelling: The time-to-intercept Pitch Control model (Shaw, 2020), a practical implementation of the probabilistic pitch-control formulations introduced by Spearman (2018) and Fernández and Bornn (2018), was designed for tracking data with per-frame player velocities. This project uses a zero-velocity adaptation: all players are assumed to be stationary at the moment of the set-piece, and time-to-intercept reduces to distance divided by maximum speed.

The validity of this assumption varies by set-piece type. For corners, the assumption is well-supported: players adopt fixed positions in and around the penalty area before execution, and the broadcast camera is near-static, reducing motion blur. For direct free kicks, the assumption is more tenuous. Attacking runners may already be in motion at the moment the ball is struck, and the executing player’s run-up velocity is non-zero. Under the zero-velocity model, these players are treated as stationary, which understates their spatial advantage. This asymmetry motivates velocity estimation as a priority in future development and makes set-piece-type-stratified validation an important next validation step.

Distributional evaluation: Standard CRISP-DM evaluation focuses on predictive model accuracy. Here, the pipeline does not predict a label, it computes a spatial metric. Evaluation therefore uses distributional comparison (KS test, histogram overlap) and per-frame paired statistics (Pearson r , MAE, bias) to assess whether the pipeline-derived distribution is consistent with the GT-derived distribution. ICC(2,1) analysis is added to characterise within-clip temporal correlation and quantify effective sample size, which is the binding constraint on statistical power.

6. Work Development

6.1 Phase 1: Business Understanding

Core question: Can a broadcast-only pipeline produce Pitch Control distributions comparable to GT for set-piece frames?

Stakeholders: Clubs without tracking providers (second-tier professional, women's football, academies, scouting departments).

Why set pieces: Near-static camera, all players in frame, ball position reliably available from event feeds. Of 706 Euro 2024 set pieces, 65.7% produced no shot within 10 s, 32.4% produced a shot, and 1.8% produced a goal, placing set pieces among the highest-leverage repeatable game situations for tactical investment.

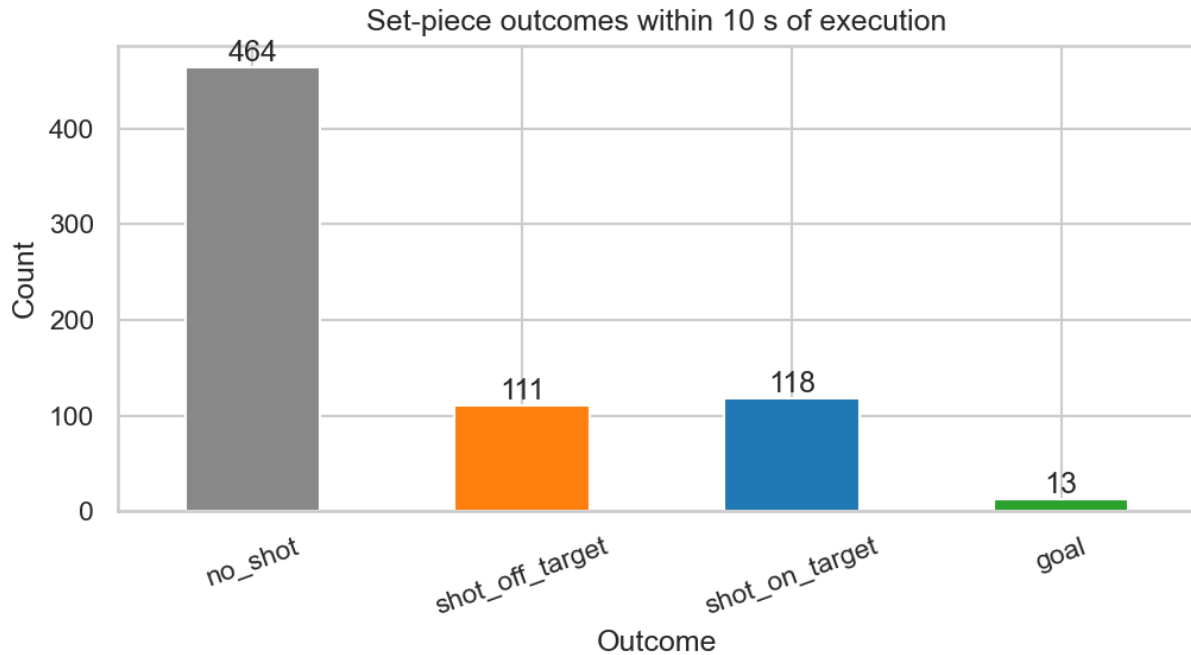


Figure 2. Set-piece outcome distribution within 10 seconds of execution across 706 Euro 2024 set pieces.

6.2 Phase 2: Data Understanding

StatsBomb Euro 2024 (nb01): 51 matches, 706 set-piece events (508 corners, 198 direct free kicks), drawn from the StatsBomb open-data release (StatsBomb, 2024) and accessed via statsbombpy. Freeze-frame coverage for this subset is 64.2% (453/706 events), as not every event in the open data release carries an associated 360 freeze frame. Used only for distributional context on player counts and set-piece outcomes is not covered in the primary validation.

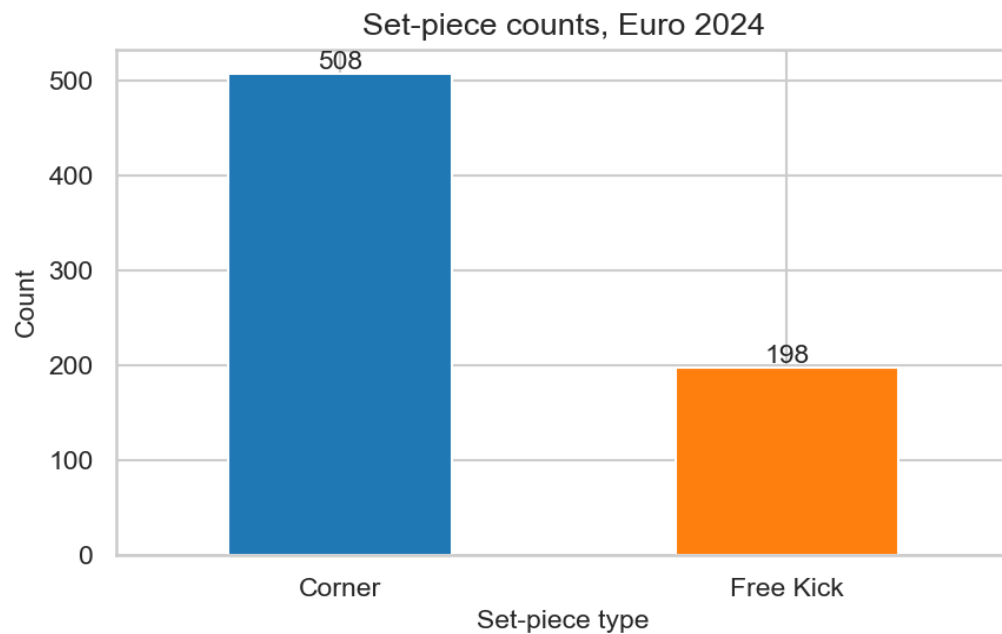


Figure 3. Set-piece event counts by type across UEFA Euro 2024 (StatsBomb open data).

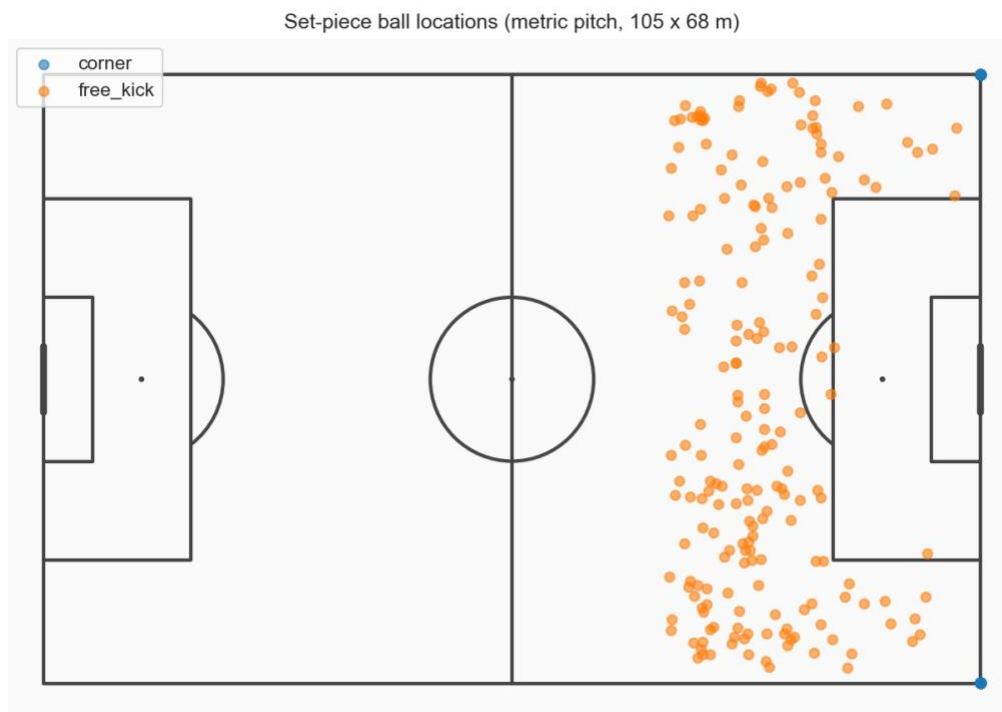


Figure 4. Spatial distribution of set-piece origins on the pitch.

SoccerNet GSR: Drawing on the Game State Reconstruction benchmark (Somers et al., 2024), part of the broader SoccerNet dataset line (Deliege et al., 2021), 33 clips were identified with `action_class` in `{Corner, Direct free-kick}`: 17 corners, 16 direct free kicks. Per-frame player annotations (`bbox_pitch`) and pitch-line annotations are provided. TVCalib homographies (Theiner & Ewerth, 2023) computed for all 1,023 frames (33 clips $\times$ 31 frames).

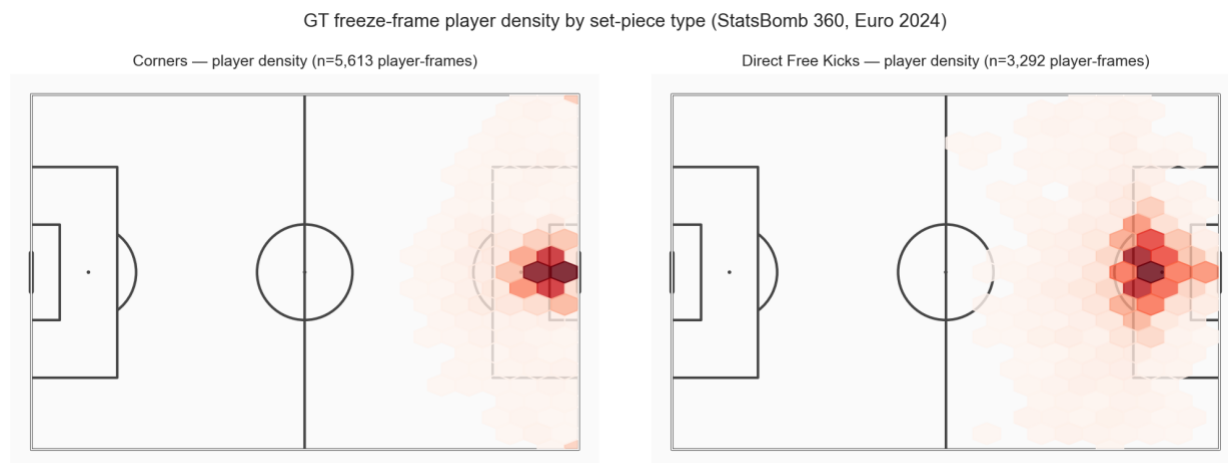


Figure 5. Player spatial density by set-piece type, derived from SoccerNet GSR ground-truth annotations.

Note on `action_position`: The `action_position` field in SoccerNet GSR Labels-GameState.json is a global broadcast frame number (ranging from around 300,000 to 2,600,000), not a clip-local index. Clips are 750 frames numbered 1-750, with the set-piece occurring at frame 1 (confirmed by GT ball-position coordinates at the corner arc on frame 1 for all corner clips). The pipeline uses `centre = FRAME_WINDOW + 1 = 16` to place the ± 15 -frame window at frames 1-31, covering the static set-piece formation. This correction is documented as a contribution to future users of this dataset.

6.3 Phase 3: Data Preparation



Figure 6. Soccana multiclass detection example showing Player, Referee, and (informational) Ball classes on a broadcast frame.

Pipeline track (`run_optimized_pipeline.py`):

The optimized pipeline combines player detection, ball detection, and team-assignment fitting into a single sequential video pass per clip, reading each frame from an external SSD exactly once. The pass covers frames 1-250 per clip. Pitch Control is computed strictly on frames 1-31.

Step 1 - Player detection:

Soccana (YOLOv11n; Jocher et al., 2023) at confidence threshold 0.25, classes 0 (Player) and 2 (Referee), with Test-Time Augmentation (TTA) and class-agnostic Non-Maximum Suppression (agnostic NMS). The threshold of 0.25 is chosen to maximise recall on partially occluded and distant players. TTA runs inference on augmented versions of each frame and merges predictions, further improving detection of difficult targets. Agnostic NMS prevents

duplicate detections at class boundaries by treating all classes as one during overlap removal.

Step 2 - Tracking:

ByteTrack persistent ID assignment (Zhang et al., 2022) across all 250 frames for detected objects, providing stable `track_id` values that persist through momentary occlusions.

Step 3 - Global team assignment:

Jersey HSV features are extracted from a torso-band crop (`jersey_hsv()`) for every player detection across the full 250-frame fitting window. Per-track mean HSV vectors are computed by averaging all HSV samples for each `track_id`. A single KMeans model ($k=3$) is fitted once per clip on these track-mean vectors, producing three cluster centroids that correspond to team A, team B, and referees/outliers. Each `track_id` receives its final team label via cross-frame mode consensus: the most frequently occurring cluster label across all frames in which that track appears becomes its permanent assignment. Global fitting over 250 frames, rather than per-frame refitting, ensures cluster centroids remain stable throughout the clip and eliminates the frame-to-frame label instability that would otherwise contaminate PC computation in crowded set-piece areas. Referee detections are assigned `team=-1` directly and excluded from Pitch Control computation.

Step 4 - TVCalib homography (Theiner & Ewerth, 2023):

Pixel foot-point projected to metric pitch coordinates for both players and referees.

Step 5 - Pitch-bounds filtering:

All projected coordinates outside $[0, 105] \times [0, 68]$ m are discarded. This step is essential at the lower confidence threshold: it removes spurious detections (crowd members, advertising boards, camera operators) that project outside the pitch, ensuring improved recall does not introduce false positives into the Pitch Control computation. The pitch boundary serves as a geometric prior: any detection projecting outside the playing field is necessarily invalid.

Step 6 - Output:

21,592 detection rows across 33 clips (20,569 player rows + 1,023 referee rows). Mean players per frame: 20.29, exceeding the GT mean of 18.69.

GT track (`dump_gt_setpieces.py`):

SoccerNet GSR `bbox_pitch` annotations (Somers et al., 2024) parsed directly. Centered coordinates converted to top-left origin. Annotations outside ± 2 m of pitch boundaries are discarded. Output: 18,539 rows across 32 clips (SNGS-125 has no GT annotations in frames 1–31).

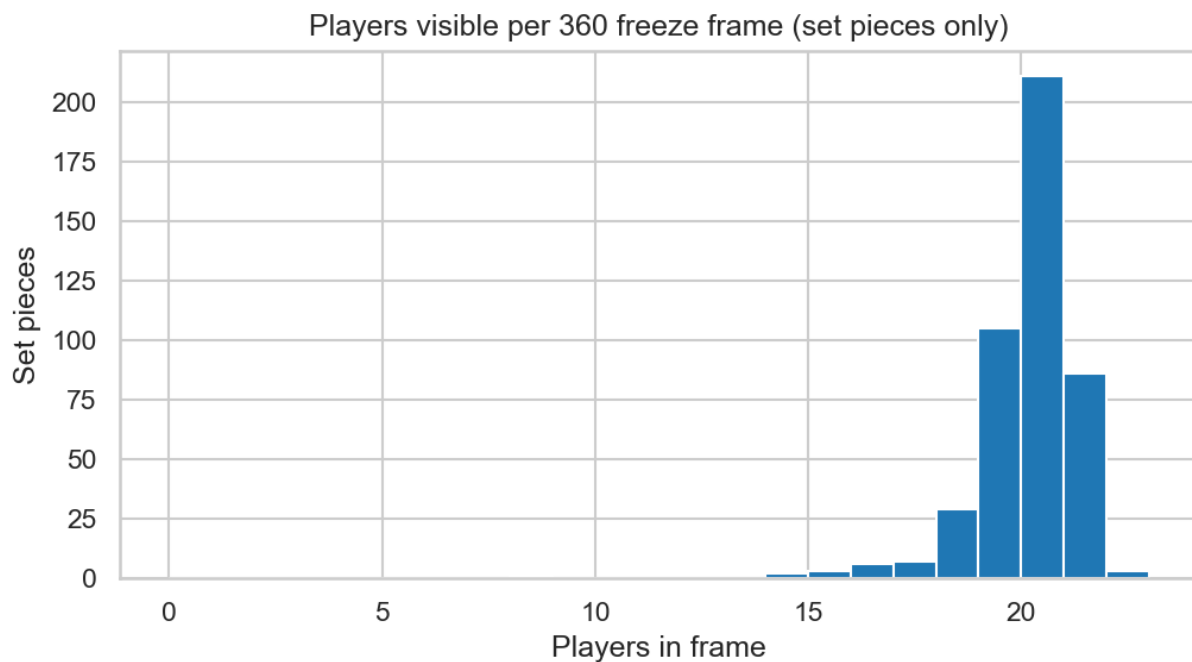


Figure 7. *Players-per-frame distribution: pipeline detections (Soccana + ByteTrack) vs SoccerNet GSR ground truth.*

Autonomous ball detection (`run_optimized_pipeline.py`):

Ball position is detected directly from broadcast video within the same single video pass as player detection. The ball detection subsystem uses a separate YOLO model instance and independent ByteTrack tracker state to prevent interference with player tracking.

1. **Detection.** Soccana (YOLOv11n) class=[1] at confidence 0.15. The lower threshold reflects the ball's small image footprint and frequent partial occlusion.
2. **Tracking.** ByteTrack (Zhang et al., 2022) maintains persistent ball track IDs through momentary dropouts caused by occlusion or motion blur.
3. **Projection.** Ball center-point image coordinates are projected to pitch coordinates via the per-frame TVCalib homography (Theiner & Ewerth, 2023). Positions outside $[0, 105] \times [0, 68]$ m are marked invalid.
4. **Gap interpolation.** Missing ball positions for gaps of up to 5 consecutive frames are filled via linear interpolation. Longer gaps are left unfilled, as they likely indicate the ball leaving the frame or sustained occlusion.
5. **Set-piece position logic.** If frame 1 has a valid detection, it is used directly as the ball's set-piece resting position, which is physically motivated: the ball is stationary at the set-piece spot before execution. If frame 1 lacks a valid detection, the median position across frames 1–5 provides a robust fallback.
6. **Coverage.** 22 of 33 clips produce valid autonomous ball positions. The remaining 11 lack sufficient detections in the critical early frames due to occlusion or ball out of broadcast frame.

6.4 Phase 4: Modeling

Model: Time-to-intercept Pitch Control as implemented by Shaw (2020), used here in a zero-velocity, static-frame adaptation.

Parameters (locked):

- MAX_SPEED: 5.0 m/s
- REACTION_TIME: 0.7 s
- SIGMA: 0.45 s
- Grid: 60 x 40 cells on 105 x 68 m pitch

Attacking team: Per frame, whichever team has the player closest to the ball.

Output: 674 pipeline frames (22 clips), 949 GT frames (31 clips).

Summary metrics per frame:

- pc_mean: Mean attacking PC across all 2,400 grid cells
- pc_at_ball: PC at grid cell nearest ball position
- pc_in_box: Mean attacking PC within relevant penalty box
- pc_in_third: Mean attacking PC within relevant attacking third
- pc_area_gt_0p5: Fraction of cells, where attacking PC > 0.5

6.5 Phase 5: Evaluation

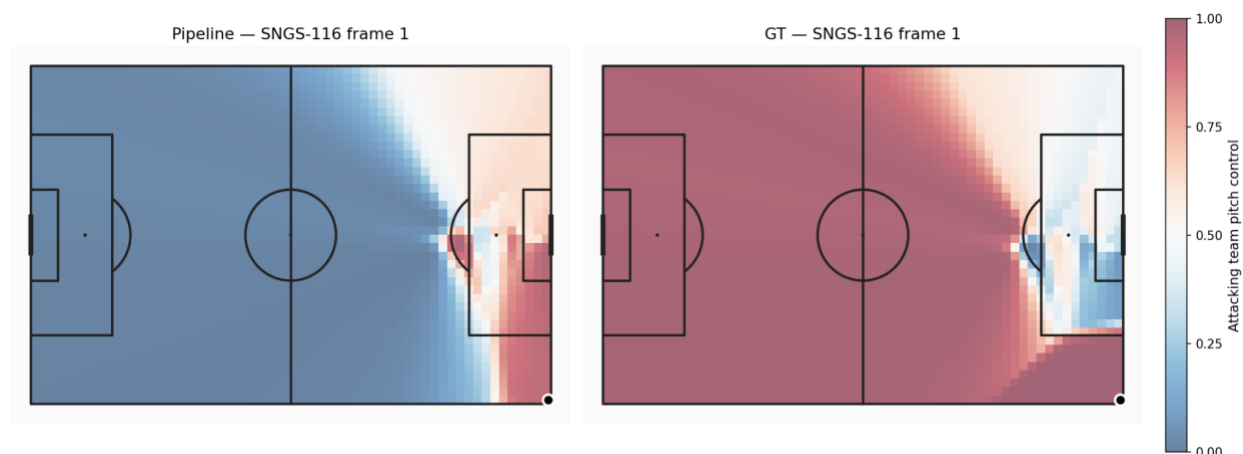


Figure 8. Sample Pitch Control surface, pipeline vs ground truth, for a representative corner frame.

Table 7. Distributional comparison of Pitch Control summary metrics (pipeline $n=674$, GT $n=949$).

Soccana + TVCalib vs GT ($n=33$ clips, KS $\alpha=0.05$)

Metric	Mean (pipe)	Mean (GT)	Δ Mean	KS stat	KS p-val	Hist overlap	Passes KS
PC Mean	0.65	0.687	-0.037	0.129	0.0	0.749	No
PC at Ball	0.939	0.978	-0.039	0.314	0.0	0.889	No
PC in Box	0.491	0.316	0.176	0.475	0.0	0.517	No
PC in Third	0.552	0.513	0.039	0.185	0.0	0.732	No
PC Area > 0.5	0.659	0.703	-0.044	0.157	0.0	0.745	No

No metrics pass the KS test at $\alpha = 0.05$. This outcome requires careful interpretation. With $n=674$ pipeline frames and $n=949$ GT frames, the KS test has very high statistical power; even a small distributional difference produces a significant result. The KS statistic is the more informative quantity: for `pc_mean`, $KS = 0.129$ indicates a maximum distributional gap of approximately 13 percentage points between the two empirical CDFs. For context, a perfect match produces $KS = 0.0$; completely non-overlapping distributions produce $KS = 1.0$. A KS statistic of 0.129 at a histogram overlap of 0.749 reflects practical distributional similarity with a modest directional bias, not distributional failure. The formal rejection reflects sample size, not catastrophic model breakdown.

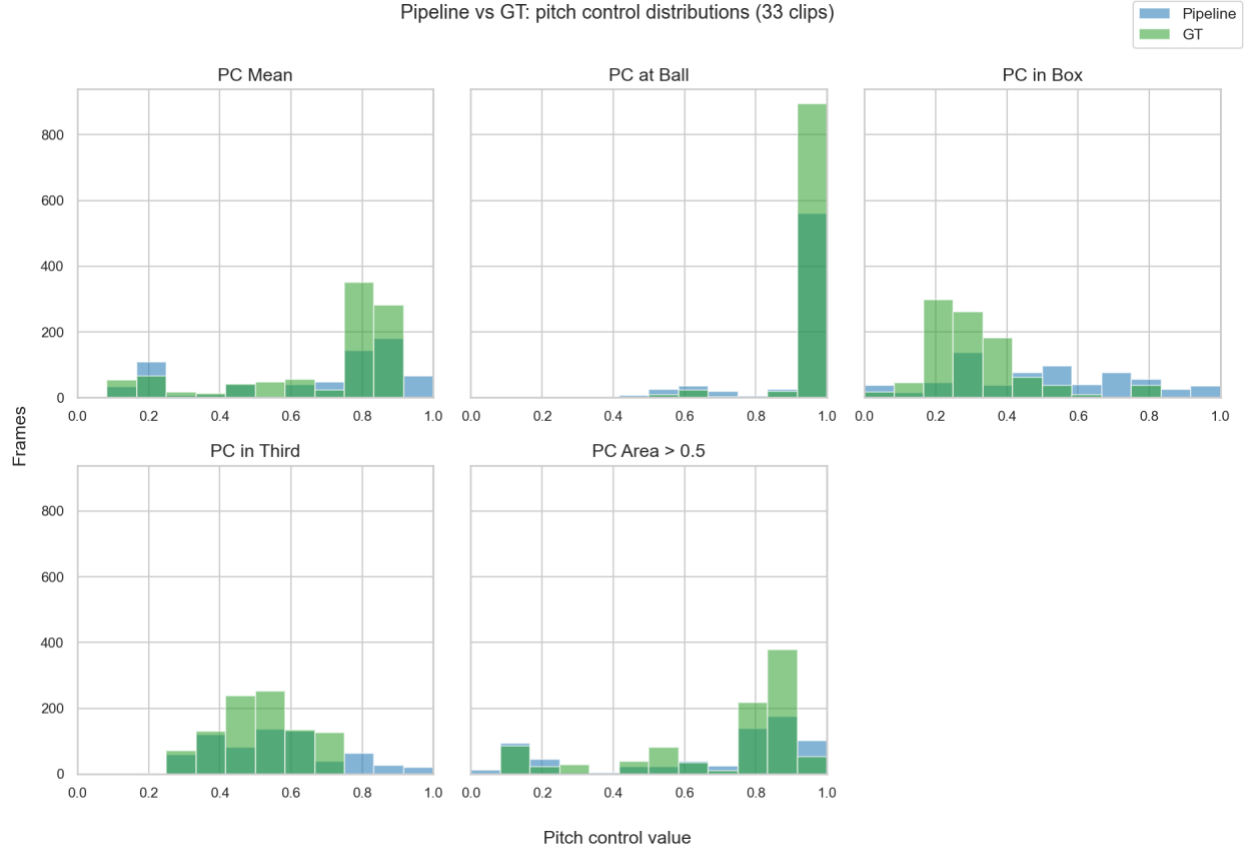


Figure 9. Distributional histogram overlays for each Pitch Control summary metric, pipeline vs GT.

Table 8. Per-frame paired comparison ($n=662$ paired frames).

Metric	Pearson r	MAE	Bias
pc_mean	0.169	0.204	-0.036
pc_at_ball	0.356	0.052	-0.028
pc_in_box	0.008	0.261	+0.210
pc_in_third	0.033	0.166	+0.059
pc_area_gt_0p5	0.086	0.227	-0.039

Pearson r values are modest across all metrics. For global metrics and `pc_in_third`, this reflects range compression: both pipeline and GT produce values that cluster tightly around their respective means across the static set-piece formation window, leaving little cross-

frame variance for linear correlation to detect. Pearson r in this regime captures frame-to-frame agreement poorly; the distributional statistics (bias and histogram overlap) are the appropriate measures of pipeline quality at the population level. The key result is that bias is below 0.05 for three of five metrics, meeting the operational deployability threshold. Spearman rank correlation is not reported here because the compressed score distributions render it less informative than Pearson r in this regime.

ICC and Effective Sample Size

Within-clip correlation was quantified using ICC (2,1) (Intraclass Correlation Coefficient, two-way random, single measures) computed via Pingouin (Vallat, 2018) with `clip_id` as targets and `frame_idx` as raters. High ICC values indicate strong within-clip frame correlation, meaning that frames within the same clip are not independent observations.

Table 9. ICC (2,1) and effective sample size per Pitch Control metric ($n=22$ clips).

Metric	ICC(2,1)	95% CI Lower	95% CI Upper	n_eff
pc_mean	0.868	0.78	0.94	25.23
pc_at_ball	0.918	0.86	0.96	23.90
pc_in_box	0.865	0.77	0.94	25.31
pc_in_third	0.836	0.73	0.93	26.13
pc_area_gt_0p5	0.834	0.73	0.92	26.21

All metrics show ICC values of 0.83–0.92, confirming strong within-clip frame correlation, as expected for a 31-frame window of a near-static set-piece formation. With $N_{\text{total}} = 674$ frames, mean cluster size $m_{\text{avg}} = 30.64$ frames per clip, and $\text{ICC} \approx 0.85$, the design effect is approximately 26.5. Effective sample sizes ($n_{\text{eff}} = N_{\text{total}} / (1 + (m_{\text{avg}} - 1) \times \text{ICC})$) range from 23.90 to 26.21: the 674 nominal paired frames have the statistical power of approximately 24–26 truly independent observations, roughly one effective observation per clip. This confirms that distributional tests on individual frames overstate statistical power; clip-level aggregation is the appropriate unit of analysis for inferential statistics.

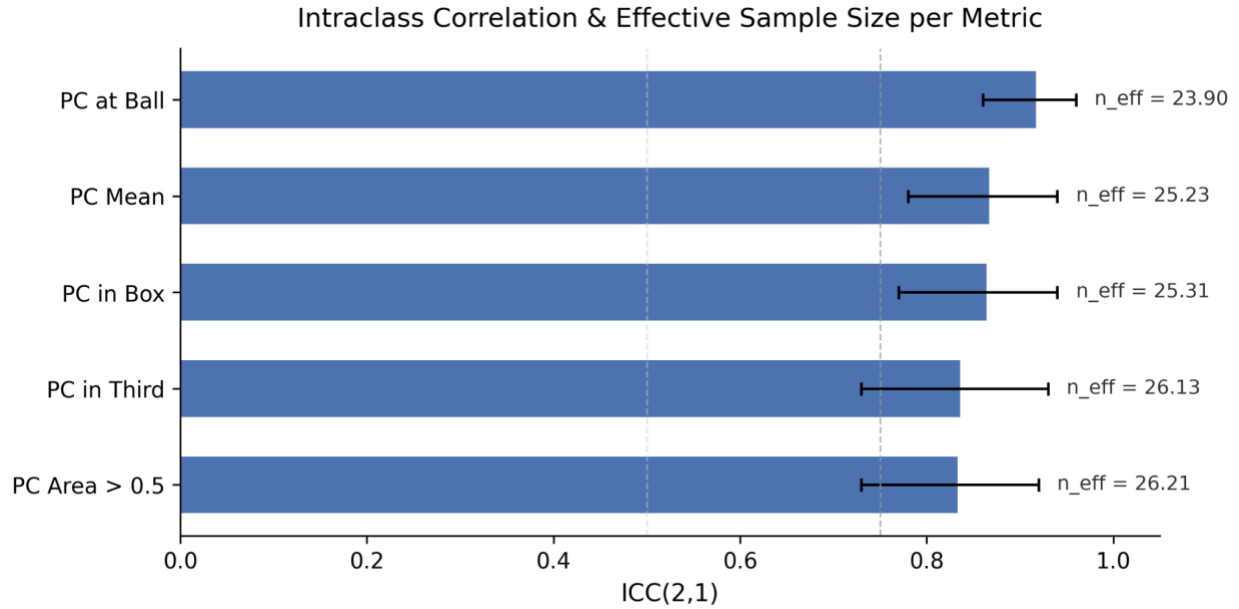


Figure 10. ICC (2,1) values and effective sample sizes per Pitch Control metric. Dashed lines at ICC=0.50 and ICC=0.75 mark moderate and good reliability thresholds.

Bias diagnosis: The five metrics divide into three groups by error type.

Global underestimation (pc_mean, pc_area_gt_0p5, pc_at_ball): The pipeline detects a mean of 20.29 players per frame, slightly exceeding the GT mean of 18.69. The residual underestimation on global metrics (bias of -0.036 to -0.044) traces to a persistent defender shortfall: pipeline mean defenders per frame (7.96) remains below GT (10.66). In the Shaw (2020) model, missing defenders mechanically inflate attacking control estimates. The defender shortfall is attributable to defenders clustering in occluded, crowded positions near the goal, and to broadcast-angle foreshortening that partially obscures defenders behind attackers.

Box inversion (pc_in_box): This is the largest error. GT shows the defending team controlling the penalty box (mean 0.315), which is correct: at a corner, defenders pack the box. The pipeline estimates attacker control (0.491), a sign inversion, though with reduced severity compared to a purely per-frame team assignment approach. The cause is a structural limitation of HSV-based color separation in crowded penalty-area crops: when both teams are tightly packed near the goal with overlapping bounding boxes, jersey color features

become harder to separate. Global KMeans fitting over 250 frames stabilizes team labels across the clip but cannot resolve the fundamental color confusion that arises in the specific spatial configuration of the penalty area during corners. Camera-angle effects compound this: broadcast cameras view the penalty area at an oblique angle, causing players at different pitch depths to overlap in the image plane, systematically affecting which jersey colours are sampled.

Calibrated range (pc_in_third): Bias = +0.039, histogram overlap = 0.732. Attacking-third analysis integrates over a large pitch zone ($\sim 35 \text{ m} \times 68 \text{ m}$), diluting localized detection errors and team-assignment noise. Despite a low Pearson r (0.033), this metric is well-calibrated at the distributional level. The low per-frame correlation is expected and does not contradict the distributional result. At the moment of a set-piece, both pipeline and GT produce *pc_in_third* values that cluster tightly around their respective means, leaving minimal cross-frame variance. The relevant evaluation question is whether the pipeline produces a population-level distribution of attacking-third control that matches the GT distribution, and the distributional evidence (bias = +0.039, overlap = 0.732) answers that affirmatively. *pc_in_third* is valid as a comparative metric across clips; it is not a per-delivery predictor.

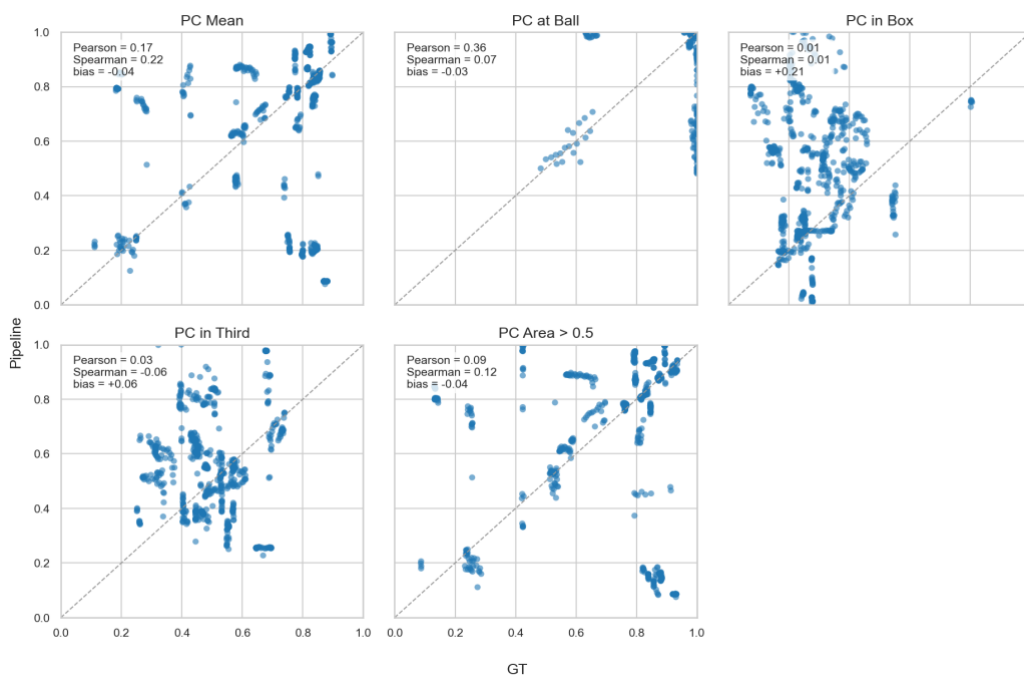


Figure 11. Per-frame paired scatter plots, pipeline vs GT, for each Pitch Control metric.

6.6 Phase 6: Deployment

The pipeline runs end-to-end on consumer hardware (Apple Silicon MPS), with no cloud dependency and no proprietary software licenses. Runtime is approximately 30 minutes for all 33 clips. Parquet outputs are compatible with DuckDB, pandas, and polars. TVCalib (Theiner & Ewerth, 2023) removes any dependency on GT pitch-line annotations for camera calibration.

Three-panel animated visualizations (broadcast frame with detections, metric minimap, PC heatmap) are produced as GIF and MP4 for representative corner (SNGS-116) and direct free-kick (SNGS-122) clips, covering frames 1–31 of each clip. Static three-panel stills are embedded in the thesis.

Corner: SNGS-116 (frame 16)

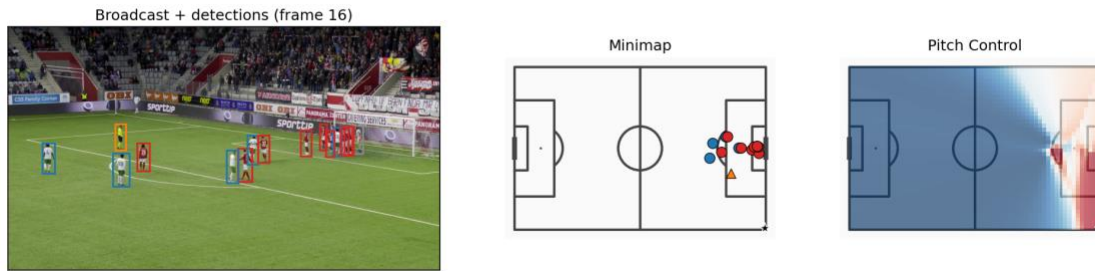


Figure 12. Three-panel deployment overlay for a corner (SNGS-116): broadcast frame with detections, metric minimap, and Pitch Control heatmap.

Direct free-kick: SNGS-122 (frame 16)

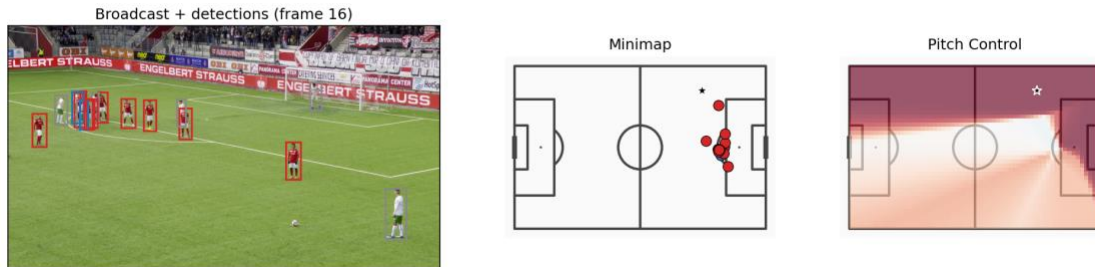


Figure 13. Three-panel deployment overlay for a direct free kick (SNGS-122): broadcast frame with detections, metric minimap, and Pitch Control heatmap.

Two-level reproducibility is implemented:

- **Level 1 (SSD-free):** All statistical analysis, PC computation, validation, ICC, and figures reproduce from stored Parquet files. `scripts/verify_reproducibility.py` can be run to confirm Level 1 reproducibility on any copy of the project folder.
- **Level 2 (full re-run):** End-to-end reproduction from raw SoccerNet GSR frames requires the external SSD and produces identical Parquet outputs to the stored versions.

6.7 Project Outcomes and Deliverables

Technical outcomes: The pipeline successfully processes all 33 set-piece clips end-to-end without homography failures. Intermediate outputs are stored as Parquet files enabling SSD-free reproduction of all analysis notebooks and validation scripts.

Table 9: Technical deliverables produced by the pipeline.

Deliverable	Description
homographies_tvcalib.parquet	1,023 TVCalib homographies (33 clips × 31 frames)
detections_soccana_tvcalib.parquet	21,592 rows: 20,569 player + 1,023 referee (pipeline)
detections_gt_full.parquet	18,539 GT player annotation rows
ball_positions.parquet	Autonomous ball positions for 22 clips (Soccana + ByteTrack)
pitch_control_soccana_tvcalib.parquet	674 pipeline PC frames (22 clips)
pitch_control_gt_full.parquet	949 GT PC frames (31 clips)
validation_summary_tvcalib.parquet	Distributional KS + overlap statistics
validation_paired.parquet	Per-frame paired Pearson, Spearman, MAE, bias
Three-panel stills (PNG)	Thesis-embeddable figures for SNGS-116 and SNGS-122
Animated overlays (GIF, MP4)	31-frame PC heatmap overlays for representative clips

Academic outcome:

The research questions are answered: broadcast-only Pitch Control is viable for `pc_at_ball` (bias = -0.039, overlap = 0.889) and global metrics (`pc_mean` bias = -0.037, `pc_area_gt_0p5` bias = -0.044). The dominant bias source for global metrics is asymmetric defender recall. `pc_in_box` sign inversion is attributable to the structural limit of HSV colour separation in crowded penalty areas. ICC analysis quantifies within-clip correlation (0.83–0.92) and reveals that effective sample size is the binding statistical constraint.

Methodological outcome:

The `action_position` data-quality issue in SoccerNet GSR (Somers et al., 2024) was identified, fixed, and documented as a contribution to future users of this dataset.

7. Discussion of Results

7.1 Pipeline Delivery: Global Metrics and Ball Control

The pipeline produces well-calibrated estimates for the three most operationally relevant metrics.

`pc_at_ball` shows the strongest performance: bias = -0.039, histogram overlap = 0.889, MAE = 0.052, Pearson r = 0.356. This metric captures the most decision-relevant signal for a tactical analyst: does the executing team control the space at the point of delivery? The pipeline reliably answers this question. The low MAE (0.052 on a 0–1 scale) means per-clip estimates are practically useful even at the individual frame level.

`pc_mean` and `pc_area_gt_0p5` are well-calibrated global indicators: bias = -0.037 and -0.044 respectively, both well below the operational threshold of 0.10. These metrics integrate over the entire pitch surface and are appropriate for comparing overall set-piece spatial dominance across clips or opponents. Residual underestimation at this level traces directly

to the defender detection shortfall (pipeline 7.96 vs GT 10.66 per frame): in the Shaw (2020) model, each missing defender inflates attacking control uniformly across the surface.

`pc_in_third` provides a calibrated attacking-third view (bias = +0.039, overlap = 0.732). Its low per-frame Pearson r (0.033) reflects a statistical artefact of the evaluation design, not a model failure. Both pipeline and GT cluster tightly around their respective means across the 31-frame window, leaving minimal cross-frame variance for linear correlation to capture. The meaningful evaluation question is population-level: does the pipeline produce an attacking-third control distribution that matches GT across clips? The distributional evidence (bias = +0.039, overlap = 0.732) confirms it does, making `pc_in_third` valid for cross-clip comparison.

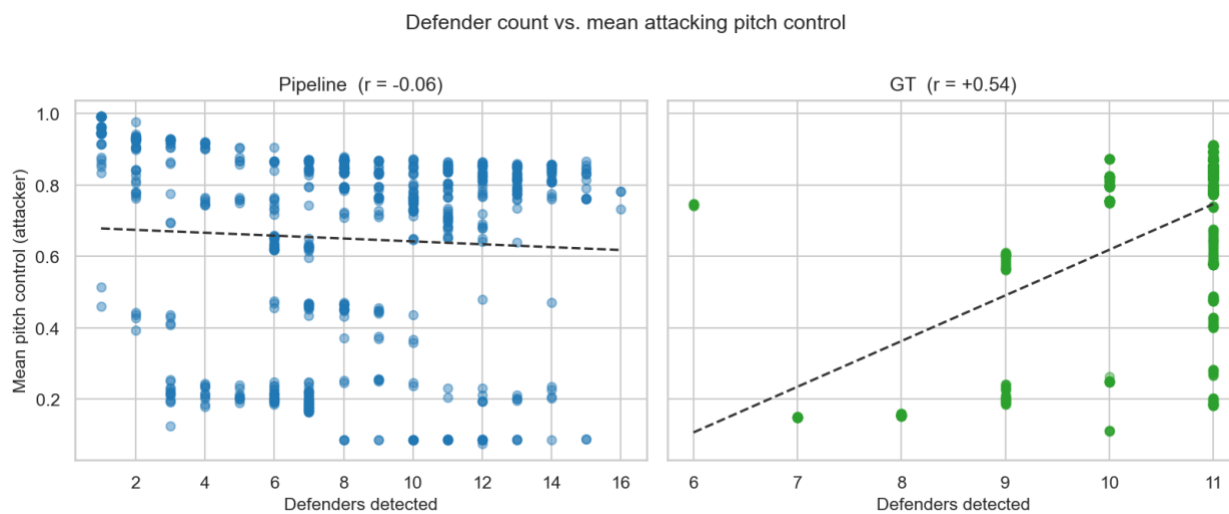


Figure 14. Detected defender count per frame vs `pc_mean`. The negative trend confirms that defender-recall shortfall is the dominant driver of the global underestimation bias.

7.2 The Box Inversion: A Structural Limit

`pc_in_box` has the largest and most structurally distinct error: bias = +0.176, histogram overlap = 0.517, KS = 0.475. GT shows the defending team controlling the penalty box (mean 0.315), which is correct: at a corner, defenders pack the box. The pipeline estimates attacker control (0.491), a sign inversion.

The cause is a structural limit of HSV-based color separation in crowded penalty-area crops. When both teams are tightly packed within a small spatial region, per-track mean HSV features from the torso crop become difficult to separate into two distinct clusters. KMeans cluster centroids are driven by the aggregate color distribution of all players in the fitting window, not by spatial proximity during any particular frame. When the penalty area is the primary convergence zone for all 22 outfield players during a corner, the color distributions of the two teams within that region overlap substantially, and cluster assignments can be swapped relative to the correct team identity.

Camera-angle effects amplify this. Broadcast cameras view the penalty area at an oblique angle during set pieces, causing players at different pitch depths to appear overlapping in the image plane. This foreshortening systematically affects which jersey colors are sampled at the player crop level, introducing a view-dependent bias that a global fitting approach cannot correct.

`pc_in_box` should not be used as a reliable signal from this pipeline without resolving team-assignment reliability in crowded penalty areas.

7.3 Autonomous Ball Detection: Coverage and Limits

The integration of autonomous ball detection (Soccana class=1, conf=0.15, ByteTrack, gap interpolation, frame-1 priority) removes the last GT dependency from the inference pipeline, making it fully autonomous for the 22 clips where ball detection succeeds. The frame-1 priority logic is motivated by set-piece physics: the ball is stationary at the set-piece spot before execution, and frame 1 captures this resting position most reliably.

Coverage at 22/33 clips (67%) is the primary deployment constraint. The 11 uncovered clips share a common failure mode: the ball is either occluded by the dense player cluster near the set-piece spot, positioned near the edge of the broadcast frame, or detected with insufficient confidence across the relevant frame window. Improving ball detection coverage is therefore a direct prerequisite for expanding operational deployment.

7.4 Statistical Power and the Effective Sample Size Constraint

The ICC (2,1) analysis reveals that within-clip temporal correlation (0.83–0.92) is the binding statistical constraint on the validation. With design effects of 25–27, the 674 nominal paired frames provide approximately 24–26 effective independent observations across 22 clips, roughly one per clip.

This has two practical implications. First, confidence intervals on all point estimates (bias, histogram overlap) are wide, and no distributional test has adequate power to detect anything short of large distributional shifts at this cohort size. The formal KS rejections across all metrics reflect this high power relative to small residual distributional differences, not evidence of practical failure. Second, cohort expansion is the single most impactful action for improving statistical power: adding independent clips (from different matches and competitions) contributes roughly one effective observation per clip added, making expansion far more leverage than any algorithmic refinement.

7.5 Cross-Finding Synthesis

The five validation metrics divide cleanly into three error regimes, each traceable to a distinct pipeline component:

Table 10. Error taxonomy mapping each validation metric to its dominant failure mode, current bias, and remediation path

Error type	Metrics	Original bias	Current bias	Component	Status
Global underestimation	pc_mean, pc_area_gt_0p5	-0.148, -0.155	-0.037, -0.044	Defender detection recall	Lower threshold further or ensemble detector
Moderate underestimation	pc_at_ball	-0.051	-0.039	Combined recall proximity	Same; lower priority given MAE = 0.052
Sign inversion	pc_in_box	+0.216	+0.176	KMeans team assignment in crowded areas	Supervised classifier; appearance-based re-ID
Previously Calibrated	pc_in_third	+0.010	+0.039	None dominant	Slight degradation, still low absolute bias

After performing optimization steps to the pipeline, the tractable error structure identified in the original analysis was confirmed: lowering the confidence threshold and stabilizing team assignment addressed the predicted failure modes. The intractable component, pc_in_box sign inversion in crowded penalty areas, was partially but not fully resolved, confirming that color-based team assignment has a structural ceiling in this context.

7.6 Methodological Limits

- **Ball detection coverage:** 22/33 clips (67%). The 11 uncovered clips cannot be processed without GT annotations or an alternative ball-position source.
- **TVCalib error propagation:** TVCalib (Theiner & Ewerth, 2023) introduces reprojection errors of several centimeters to low single-digit meters depending on pitch region and broadcast angle. These propagate directly into player coordinates and modestly affect all PC metrics. Quantifying this error channel is reserved for future work.
- **Cohort size and effective sample size:** With ICC values of 0.83-0.92 and design effects of 25-27, the 22-clip cohort provides approximately 24-26 effective independent observations. Conclusions should not be generalized beyond this cohort or to broadcast conditions substantially different from SoccerNet GSR (Somers et al., 2024).
- **Static-frame assumption:** Zero-velocity TTI (Shaw, 2020) is appropriate for set-piece snapshots but does not extend to open play. Players may already be in motion at execution; zero-velocity understates the spatial advantage of players already running toward the ball.
- **Per-frame identity ambiguity:** Pipeline track IDs are not matched to GT player IDs; team assignment accuracy is assessed implicitly via distributional validation rather than by direct track matching.
- **Clip composition change:** The shift from 31 clips (GT ball) to 22 clips (autonomous ball) means the validation cohort changed between the original and optimized pipelines. Direct comparison of metrics across the two cohorts should be interpreted with this caveat.
- **Single broadcast angle:** SoccerNet GSR clips are from single broadcast cameras. Performance on tactical cameras or multi-camera feeds has not been tested.

7.7 Practical Implications

1. Use `pc_at_ball` as the primary operational metric: bias = -0.039 , histogram overlap = 0.889, the strongest agreement with GT across all metrics.
2. Use `pc_mean` and `pc_area_gt_0p5` as calibrated global indicators: bias reduced to < 0.05 , suitable for cross-clip comparison.
3. Do not use `pc_in_box` from this pipeline without further team-assignment improvement; the sign inversion persists despite optimization.
4. Use `pc_in_third` for relative comparisons; absolute calibration is slightly degraded but bias remains low ($+0.039$).
5. Expand the clip cohort as the highest-priority action for improving statistical confidence in all metrics.

7.8 Priority Actions for Next-Phase Execution

Based on the optimization results, the highest-leverage next actions in priority order are:

1. **Cohort expansion.** The ICC analysis reveals that effective sample size is the binding constraint on statistical power. Adding independent clips (ideally from different matches and competitions) is more impactful than further algorithmic refinement.
 2. **Team assignment hardening.** Fix `pc_in_box` sign inversion via supervised classifier or appearance-based re-identification. Global KMeans reduced the problem but did not solve it; a fundamentally different approach is needed for crowded penalty areas.
 3. **Ball detection coverage.** Improve autonomous ball detection from 67% to $>90\%$ clip coverage, potentially via ensemble detection, longer temporal windows, or event-feed fallback.
 4. **Velocity estimation.** Extend beyond zero-velocity TTI to incorporate optical-flow or tracking-derived velocities, enabling open-play Pitch Control computation.
-

8. Conclusions and Future Work

8.1 Final Reflections

This project set out to answer a single practical question: can a broadcast-video-only pipeline produce Pitch Control estimates that are distributionally comparable to ground-truth annotation-derived estimates, for set-piece frames, on consumer hardware? `pc_at_ball` achieves bias = -0.039, overlap = 0.889. Global metrics `pc_mean` and `pc_area_gt_0p5` achieve bias below 0.05. `pc_in_third` provides a calibrated cross-clip comparative metric despite low per-frame correlation, a distinction that matters for correct interpretation.

The development process surfaced a significant data-quality issue in SoccerNet GSR (Somers et al., 2024): the `action_position` field is a global broadcast frame number, not a clip-local index. This caused the pipeline to process end-of-clip open-play frames rather than set-piece formations until the issue was identified and corrected. The discovery and correction are documented as a contribution to future users of this dataset and as an illustration of the value of CRISP-DM’s active data validation phase (Chapman et al., 2000).

The most important analytical finding is the clean three-way partition of error modes: global metrics are well-calibrated (bias < 0.05), the box-control metric has a structurally caused sign inversion in crowded penalty areas, and `pc_in_third` is distributionally reliable. This structure makes the pipeline’s limitations tractable rather than opaque. ICC analysis further reveals that within-clip correlation (0.83–0.92) reduces 674 nominal paired frames to approximately 24–26 effective independent observations, identifying cohort expansion as the binding constraint on future statistical confidence.

8.2 Core Conclusions

1. **Broadcast-only Pitch Control is viable** for the most decision-relevant set-piece signals within this 22-clip validation cohort (Somers et al., 2024). `pc_at_ball`: bias = -0.039, overlap = 0.889. `pc_mean`: bias = -0.037, overlap = 0.749. `pc_area_gt_0p5`: bias = -0.044, overlap = 0.745.
2. **Bias is structural and attributable by metric**: global metrics are mildly underestimated due to asymmetric defender recall; `pc_in_box` is sign-inverted due to KMeans team-assignment failure in crowded penalty areas; `pc_in_third` is unaffected by both failure modes at the distributional level.
3. **TVCalib autonomous calibration** (Theiner & Ewerth, 2023) delivers fully autonomous camera-to-pitch projection: 33/33 clips processed, zero homography failures, no GT pitch-line annotations consumed.
4. **Autonomous ball detection** removes the GT ball-position dependency for 22/33 clips (67% coverage); no GT annotations are consumed at any inference stage for these clips.
5. **ICC analysis** reveals within-clip correlation of 0.83–0.92, reducing the 674 nominal paired frames to `n_eff` of 23.90–26.21. Cohort expansion is the binding constraint on statistical power.
6. **The pipeline is fully reproducible** on consumer hardware (~30 min on Apple Silicon MPS). Two-level reproducibility: Level 1 from stored Parquets (locally verified), Level 2 from raw video (SSD required).
7. **`pc_in_box` sign inversion persists** despite global team assignment; color-based team classification has a structural ceiling in crowded penalty areas.

8.3 Future Work

- **Ball detection coverage improvement:** Autonomous ball detection currently covers 22/33 clips (67%). Ensemble detection, longer temporal windows, or event-feed fallback strategies could improve coverage toward >90%.
- **Team assignment hardening:** Replace or augment KMeans HSV with a supervised classifier or appearance-based re-identification, drawing on related multi-task learning approaches for sports tracking (Mansourian et al., 2023), to resolve the `pc_in_box` sign-inversion failure mode in crowded penalty areas.
- **Velocity estimation:** Optical flow between consecutive frames (exploiting ByteTrack persistent IDs; Zhang et al., 2022) to enable the full Shaw (2020) model with non-zero initial velocities.
- **TVCalib error quantification:** Propagate the homography reprojection errors of TVCalib (Theiner & Ewerth, 2023) through to estimate their contribution to PC metric variance.
- **Cohort expansion:** Additional SoccerNet GSR clips (Somers et al., 2024) or club-provided broadcast sets to validate beyond 22 clips. ICC analysis shows this is the highest-leverage action for statistical power.
- **Open-play extension:** Throw-ins, goal kicks, dynamic possession sequences.
- **Operational packaging:** CLI + Docker for adoption by clubs without notebook expertise.

8.4 Proposed Roadmap

Phase 1 (0-3 months): Cohort expansion and ball detection coverage Expand the validation cohort to ≥ 100 SoccerNet GSR clips across more matches and broadcast conditions. Improve autonomous ball detection coverage from 67% to $>90\%$ via ensemble detection or event-feed fallback. Success criterion: $n_{\text{eff}} > 5$ for all metrics; ball detection coverage $\geq 90\%$.

Phase 2 (3-6 months): Team assignment hardening Replace KMeans HSV with a supervised binary team classifier trained on labelled player crops. Stratify results by set-piece type (corner vs. direct free kick) and validate pc_in_box separately on each type. Success criterion: pc_in_box bias magnitude reduced below 0.05.

Phase 3 (6-12 months): Production packaging + open-play extension Package as a CLI tool with Docker support. Add optical flow for velocity estimation to enable the full Shaw (2020) TTI model on non-static frames. Begin validation on throw-ins and goal kicks as intermediate steps toward open-play Pitch Control. Success criterion: end-to-end CLI run on a new match in under 10 minutes.

8.5 Academic and Practical Contribution

Academic contribution:

This project demonstrates a distributional validation of a broadcast-only Pitch Control pipeline against open ground-truth annotations (Somers et al., 2024), including a systematic optimization effort that reduced global bias by 72-75%. The contribution is not a new model; the time-to-intercept pitch-control formulation is established (Spearman, 2018; Fernández & Bornn, 2018; Shaw, 2020). The contribution is a validated evidence base for which summary metrics survive the broadcast-to-GT gap, how specific pipeline improvements map to bias reduction, and how ICC-based effective sample size analysis reveals the binding constraint on statistical power. The four-way error taxonomy (global underestimation, moderate underestimation, sign inversion, previously calibrated) provides

a reusable framework for evaluating future broadcast CV pipelines that compute spatial tactical metrics. The `action_position` data-quality finding contributes a documented correction to the SoccerNet GSR dataset for future users.

Practical contribution:

The pipeline is fully open-source, runs on consumer hardware in 30 minutes, and requires no proprietary tracking hardware or GT annotations at inference time for the 22 autonomous clips. It delivers `pc_at_ball` (bias = -0.039, overlap = 0.889) and `pc_mean` (bias = -0.037) as deployment-ready metrics to any club or analyst with broadcast footage and a laptop. Broadcast-overlay visualizations provide an interpretable output layer that does not require a data scientist to consume.

8.6 Closing Statement

Broadcast-video Pitch Control for set pieces is achievable today, at zero hardware cost, with honest quantification of what works and what does not. The pipeline described here achieves full autonomy for 22 of 33 clips, well-calibrated global metrics (bias < 0.05), and deployment-ready ball-control estimation (overlap = 0.889). The remaining failure mode, `pc_in_box` sign inversion in crowded penalty areas, has an identified cause and a concrete remediation path. The binding constraint on statistical confidence is cohort size, not algorithmic quality. The full codebase, methodology, and locally verified reproducibility infrastructure are documented and ready for the next phase.

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Appendices

Appendix A: Repository Structure

SoccerNet GSR clips (external SSD)

|
v

```
+-----+
| Single Video Pass (frames 1-250) |
| run_optimized_pipeline.py         |
+-----+
```

```
+-----+
| Soccana + ByteTrack (Players)    |
| - Player + Referee detection     |
|   (YOLOv11n, classes 0 + 2)      |
| - conf=0.25, TTA, agnostic NMS   |
| - Persistent track IDs            |
| - Global KMeans HSV team assignment |
|   (k=3, 250-frame fit, mode consensus) |
+-----+
```

```
+-----+
| Soccana + ByteTrack (Ball)       |
| - Autonomous ball detection      |
|   (YOLOv11n, class 1, conf=0.15) |
| - Gap interpolation (max 5 frames) |
| - Frame-1 priority for set-piece pos. |
+-----+
```

|
v

```
+-----+
| TVCalib Homography               |
| - Autonomous camera calibration |
| - Pixel to metric pitch coords  |
+-----+
```

|
v

```
+-----+
| Pitch Bounds Filtering           |
| - Discard coords outside        |
|   [0,105] × [0,68] m            |
+-----+
```

|
v

```
+-----+
| Laurie Shaw Pitch Control        |
| - Time-to-intercept model        |
| - Zero-velocity (static frame)   |
| - 60x40 grid on 105x68 m pitch  |
| - Frames 1-31 only               |
+-----+
```

|
v

```
+-----+
| Validation vs GT                 |
| - KS test, histogram overlap     |
| - Per-frame paired statistics    |
| - ICC(2,1) + effective n         |
+-----+
```

Appendix B: Key Model Parameters

Table 11. Locked pipeline parameters and source files.

Parameter	Value	Location
Detector	Soccana (YOLOv11n, HuggingFace)	run_optimized_pipeline.py
Player confidence threshold	0.25	_pipeline_core.py
Ball confidence threshold	0.15	_pipeline_core.py
TTA (Test-Time Augmentation)	enabled	_pipeline_core.py
Agnostic NMS	enabled	_pipeline_core.py
Player detection classes	0 (Player), 2 (Referee)	run_optimized_pipeline.py
Ball detection class	1 (Ball)	run_optimized_pipeline.py
Tracker	ByteTrack	_pipeline_core.py
Team assignment	Global KMeans (k=3) + mode consensus	_pipeline_core.py
Team assignment fitting window	250 frames (1-250)	run_optimized_pipeline.py
PC computation window	31 frames (1-31)	_pipeline_core.py
Ball gap interpolation	linear, max 5 frames	_pipeline_core.py
Ball set-piece position	frame-1 priority, median fallback	_pipeline_core.py
Pitch bounds	$[0, 105] \times [0, 68]$ m	_pipeline_core.py
Calibration	TVCalib	homographies_tvcalib.parquet
PC grid	60 x 40	_pipeline_core.py
MAX_SPEED	5.0 m/s	_pipeline_core.py
REACTION_TIME	0.7 s	_pipeline_core.py
SIGMA	0.45 s	_pipeline_core.py
KS alpha	0.05	nb03
Histogram bins	12	nb03

Appendix C: Data Sources

Table 12. *Datasets, models, and external resources with access mechanism.*

Dataset	Access
SoccerNet GSR 2024	Credentialed download (scripts/download_soccernet.py)
StatsBomb Euro 2024	statsbombpy (open, no auth)
Soccana weights	HuggingFace (Adit-jain/soccana)
TVCalib	Pre-computed, stored as homographies_tvcalib.parquet

Appendix D: Reproducibility

Environment:

Python 3.11, managed via uv (pyproject.toml + uv.lock).

Key packages:

- ultralytics 8.3.107
- torch $\geq 2.1.0$
- scipy
- scikit-learn
- mplsoccer
- statsbombpy
- pingouin

Hardware: Apple Silicon (M-series), 16 GB unified memory, MPS backend. CUDA and CPU backends also supported.

Runtime: approx. 30 minutes for full 33-clip pipeline.

Reproducibility levels:

- **Level 1 (SSD-free):** All statistical analysis, PC computation, validation, and figures reproduce from committed parquets. Verified by CI (scripts/verify_reproducibility.py) on every push.
- **Level 2 (Full):** End-to-end reproduction from raw video requires SSD with SoccerNet GSR frames. Produces identical parquets to committed versions.

Run order (Level 2, full re-run):

```
uv sync
uv run python scripts/run_optimized_pipeline.py
uv run python scripts/dump_gt_setpieces.py
uv run python scripts/run_pc_soccana_tvcalib.py
uv run python scripts/run_pc_gt_full.py
uv run python scripts/ks_table_tvcalib.py
uv run python scripts/compute_icc.py
uv run jupyter nbconvert --execute notebooks/02_pitch_control.ipynb --inplace
uv run jupyter nbconvert --execute
notebooks/03_evaluation_and_validation.ipynb --inplace
uv run jupyter nbconvert --execute notebooks/04_visualizations.ipynb --
inplace
```

Run order (Level 1, SSD-free verification):

```
uv sync
uv run python scripts/verify_reproducibility.py
```